

Title

Multimodal Nonpharmacologic Delirium Prevention Reduces Postoperative Delirium in Older Adults: A Stepped-Wedge Trial

Abstract

Postoperative delirium in older surgical patients remains a significant quality and safety challenge despite available prevention strategies. We evaluated a comprehensive, team-based delirium prevention program implemented across 5 hospitals in 1470 patients aged 70 or older undergoing elective orthopedic, general, or cardiac surgery. The multimodal program included staff education, patient assessment of delirium risk factors, and individualized non-pharmacologic interventions (cognitive engagement, sensory optimization, mobility promotion, sleep support). Overall, the intervention reduced postoperative delirium incidence by approximately 13% compared to standard care, with greater absolute reductions among patients undergoing orthopedic or general surgery (effect most pronounced for these procedure types). Delirium duration--days spent delirious during hospitalization--was also significantly reduced in the intervention group. The prevention program did not produce significant delirium reduction in cardiac surgery patients. No serious adverse events related to the prevention protocol were documented. Implementation was feasible across diverse hospital settings with appropriate staff training. These findings support adoption of multimodal delirium prevention programs, particularly for older patients undergoing orthopedic and general surgical procedures.